# TJ Weldy

**Lead Data Engineer | Data Architect**

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Open to domestic and international relocation

## SUMMARY

Lead data engineer and data architect with 20+ years across software, database, and data-platform engineering. Hands-on technical leader who designs from first principles and works at every layer—from database internals and application code to architecture and production incident response. Created two original CDC architectures: a compressed, changeset-based audit platform with cross-table history, and a self-healing pipeline framework that rebuilds itself through schema drift while preserving full change history across a 1.5B+ row log. At eBay Mobile, architected a real-time telemetry platform sustaining 30,000 inserts per second with sub-minute alerting and enabling incident response that protected millions in GMV. Now leads two DBAs and mentors cross-functional engineers, turning difficult systemic failures into durable fixes that can be scaled across an organization. Brings a QA-rooted instinct for data correctness and a record of making undocumented, business-critical systems understandable, reliable, and operable.

## CORE COMPETENCIES

**Data Platforms & Pipelines** — PostgreSQL, SQL Server, Oracle 8i–19c, DuckDB, MotherDuck, dbt, Airflow, and Kafka; real-time data pipelines, CDC, ETL/ELT, dimensional modeling, and data warehousing

**Database Engineering** — Schema and data-model design, indexing, stored-procedure optimization, execution-plan analysis, trigger- and snapshot-based CDC, autovacuum tuning, PostgreSQL major-version upgrades, and capacity and performance engineering

**Cloud & Infrastructure** — AWS RDS, Aurora PostgreSQL, EKS, S3, SQS, IAM, Parameter Store, and CloudWatch; Kubernetes, Docker, infrastructure economics, and production operations

**Languages** — SQL, PL/pgSQL, T-SQL, Python, C#/.NET, Go, Rust, and Bash; earlier professional experience with Java and Scala; TypeScript and JavaScript maintenance and debugging

**AI & Machine Learning** — Elasticsearch ML anomaly detection, self-hosted LLM inference with Ollama, GPU/CPU model tiering, agent and tool integration, and AI-assisted engineering workflows

**Delivery & Operations** — GitHub Actions, Git, 24/7 database incident response, SOC 2 Type II evidence automation, pg_stat_statements, CloudWatch, Grafana, and Alloy

# EXPERIENCE

## Architect / Lead Data Engineer

**HighRadius, following acquisition of Ascend Software**
2025–Present

- Lead two DBAs and provide hands-on technical direction to a cross-functional development team. Scope epics according to engineers' experience, develop repeatable troubleshooting guidance, mentor developers, and make direct fixes across C#, Go, TypeScript, Python, and database code.
- Inherited and stabilized systems previously owned by the departing lead architect. Serve as level-two support in the 24/7 database incident-response rotation and handle direct client escalations involving database health, performance, and legacy on-premises deployments.
- Designed and built a metadata-driven, self-healing CDC pipeline from PostgreSQL and SQL Server into MotherDuck. A trigger and code-generation framework produces Airflow-deployed DAGs that move transactional data through raw, time-variant staging, and current-record marts.
- Made the pipeline resilient to schema drift by automatically rebuilding triggers after DDL changes. The PostgreSQL implementation captures row-level changes and emits schema-change notifications to downstream consumers, preserving complete history across a 1.5B+ row change log for as-of-date analysis.
- Diagnosed a fleet-wide PostgreSQL connection pileup involving hundreds of sessions stalled on unread results. Traced the issue to application paths that failed to consume `RETURNING` results and affected-row counts, built a ruleset to identify occurrences across the codebase, and structured the remediation for distributed implementation by the development team.
- Built a dbt-duckdb Python model with adaptive, size-based batching to process a 5.7 TB data backlog within MotherDuck's 22 GB per-query memory limit. The previous single-statement approach exhausted memory within seconds; the adaptive implementation processes batches exceeding 100M events reliably.

- Identified approximately $80,000 in annual RDS rightsizing opportunities across a 13-instance SQL Server fleet through 30-day CloudWatch analysis and uncovered a Reserved Instance coverage gap in the PostgreSQL environment.
- Authored a PostgreSQL 13-to-16 major-version upgrade plan covering production and nonproduction parameter-group topology, per-table autovacuum settings for 1.5B-row tables, compatibility validation, and staged rollout sequencing.
- Led database evidence collection and audit-facing presentation for SOC 2 Type II, including automated evidence acquisition, access reviews, and query-level compliance mapping.

## Data Engineer, promoted from SDET

**Ascend Software**
2019–2025

- Progressed from SDET into a full data-engineering role through repeated production database interventions involving query rewrites, index and stored-procedure optimization, execution-plan analysis, data validation, and race-condition debugging.
- Designed a changeset-based SQL Server audit and CDC platform that stores compressed JSON differences rather than full-row snapshots, substantially reducing storage requirements. Encoded parent-child relationships allow retrieval of changes to a record, its dependent records, or the complete cross-table history of a user's activity.
- Delivered the platform as a user-facing audit trail that shows what each user changed, when the change occurred, and how the data changed.
- Developed trigger-based CDC and ETL tooling that eliminated the need for licensed third-party change-capture products.
- Prototyped a vendor-neutral ERP integration layer that maps differing ERP data structures and domain terminology into a single canonical model without requiring vendor-specific database objects.
- Built database migration tooling for SQL Server-to-PostgreSQL migrations and transfers across security boundaries using AWS RDS and AWS Database Migration Service.
- Created end-to-end data-integrity tooling spanning a PostgreSQL processing database, SQL Server customer databases, and the Workday API. The system dynamically generated financial documents, exercised production workflows, and validated results at each stage.
- Migrated CI/CD workflows from TeamCity to GitHub Actions and AWS CodeDeploy, trained developers on the new processes, and authored supporting documentation.

## Family Medical Leave

2018–2019

Full-time care for a family member during a serious illness.

### Software Engineer

**eBay Mobile**
2012–2018

- Architected eBay Mobile's real-time error and performance telemetry platform on an appliance-class Oracle system. A 100-queue ingestion layer with modulo-sharded load distribution fed a purpose-built dimensional model while sustaining 30,000 inserts per second.
- Delivered device-to-database latency below 30 seconds and device-to-alert latency below one minute while simultaneously serving multi-hour analytical aggregations in 10–20 seconds.
- Integrated Hadoop, Kafka, and Cassandra data into operational reporting and eBay's central data warehouse.
- Built anomaly-detection capabilities using Java, Scala, and Elasticsearch ML that identified degrading response times at the individual service-instance level and predicted failures before they became outages.
- Designed domain-aware alerting that distinguished regional mobile-carrier disruptions from failures within eBay infrastructure. The platform became reliable enough that eBay's Network Operations Center detected incidents through it before receiving alerts from its existing monitoring systems.
- Enabled precise incident triage during revenue-impacting service degradations, protecting millions of dollars in gross merchandise volume on multiple occasions.
- Served as a crisis-management database specialist during major incidents and developed zero-downtime deployment patterns for database schema changes.

### Earlier QA & Build Engineering Roles

**Corillian · Earth Class Mail · WebMD · Press Ganey**
Prior to 2012

Progressive roles in software quality, test automation, build engineering, CI/CD, and ETL validation across Teradata and SQL Server environments. This work established the data-quality and failure-analysis discipline that became foundational to later database and data-platform engineering roles.

## SELECTED PROJECTS

### Self-Hosted LLM Platform and Long-Running Agent

**Personal project · Ongoing**

- Design and operate a multi-model local inference environment using Ollama and Qwen2.5 14B, 32B, and 72B models, with tiered GPU and CPU placement on consumer hardware.
- Built a long-running agent process managed through systemd, including liveness heartbeats, automatic recovery, graceful time-budgeted shutdown, and persistent local state.
- Diagnosed an asynchronous event-loop starvation issue caused by two large models competing for a single GPU.
- Developed a decoupled file-bridge architecture, append-only JSONL data stores, encrypted offsite backups, and a plugin-based interface running on an embedded touchscreen.

### pg_access_layer

**Open-source PostgreSQL extension**

Developed a PL/pgSQL and PGXS extension that generates version-tracked, read-only access layers over source schemas. Features include JSON Schema column contracts, complete view-definition history, and event-trigger-based lifecycle management. Continuous integration passes on PostgreSQL 16, 17, and 18.

github.com/tweldy/pg_access_layer

### mystic-light

**Rust · Personal project**

Authored a Rust crate that controls MSI Mystic Light JARGB fan lighting directly through USB HID after reverse-engineering the device's wire protocol.

## EDUCATION

**Master of Science, Computer Information Systems**
Database Management and Business Intelligence
Boston University, 2012

**Bachelor of Arts, Economics**
Reed College